

PROFESSIONAL SUMMARY

Innovative and results-driven Software Engineer with a master's in computer science and practical experience in developing scalable, secure, and responsive full-stack web applications. Skilled in AngularJS, Spring Boot, REST API development, CI/CD automation, cloud infrastructure (AWS), and secure coding practices. Adept in applying SDLC methodologies and test-driven development to deliver high-quality, maintainable software solutions aligned with business goals. Passionate about building efficient systems with strong knowledge of design patterns, version control, and Agile methodologies.

SKILLS

- **Programming Languages:** Java, TypeScript, Python, JavaScript, SQL
 - **Frameworks & Tools:** AngularJS, React.js, Spring Boot, Next.js, Node.js, Express.js, Flask, WordPress, Supabase
 - **Testing:** JUnit, Jest, Pytest, Postman, Manual & Integration Testing
 - **DevOps & Infrastructure:** Docker, Git, GitHub, CI/CD (GitHub Actions), SonarQube, Checkmarx, VS Code, Azure Data Studio
 - **Cloud & Platforms:** AWS (EC2, S3, CloudFront, Lambda), Salesforce, Firebase, Snowflake
 - **Databases:** MySQL, MongoDB, Oracle, PostgreSQL, AWS S3
 - **Security & Architecture:** OWASP, Secure Coding, API Security, REST APIs, Microservices, SOA, SDLC (Scrum, Kanban, SAFe)
 - **Design Patterns:** Singleton, Factory, Adapter, Observer, Strategy, IoC
 - **Soft Skills:** Problem-Solving, Team Collaboration, Communication, Analytical Thinking, Adaptability
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PROJECT

Government Fund Allocation and Tracking System Using Blockchain

- Developed a blockchain-based system to improve transparency and accountability in public fund disbursement and tracking, reducing the risk of corruption and mismanagement.
- Implemented **Ethereum blockchain** and **smart contracts** using **Solidity** to ensure secure, immutable, and decentralized tracking of funds across different government levels.
- Built a **real-time web interface** for citizens and auditors to view live updates on fund allocation and utilization, enhancing public trust and engagement.
- Integrated **MetaMask** wallet for secure user authentication and transaction approval, ensuring end-to-end cryptographic security for fund flow actions.
- Built frontend with React and deployed smart contracts on Ropsten testnet, Followed secure coding practices (OWASP) and used Git for version control.
- Applied Agile methodology and used containerized environments for scalable testing.

Customer Segmentation

- Developed a customer segmentation solution using **K-means clustering** to group customers based on behavioral and demographic attributes for business insight generation.
- Preprocessed the dataset using **Python (Pandas, NumPy, Scikit-learn)** by handling missing values, encoding categorical data, and applying normalization for improved model performance.
- Utilized **Tableau** and **Seaborn** for detailed visualizations of segmented groups, revealing key trends in purchase frequency, geographic distribution, and customer value.
- Integrated clustering results into business workflows to support **targeted marketing, personalized recommendations**, and improved campaign performance.
- Integrated clustering models into backend APIs and optimized SQL queries for performance. Followed Scrum for iterative development and ensured service-level testing.

Simultaneous Localization and Mapping (SLAM) System

- Developed a monocular Visual SLAM system from scratch to estimate a moving camera's 3D trajectory using video input or sequential image frames.
- Implemented ORB feature extraction and brute-force Hamming matcher to detect and match key points between frames, enabling visual odometry.
- Designed a smart keyframe selection algorithm to identify and store only significant frames based on motion thresholds (translation and rotation), optimizing computational efficiency and trajectory accuracy
- Built a modular pipeline with separate components for feature extraction, pose estimation, keyframe selection, and trajectory visualization.
- Applied containerization for reproducible testing and CI pipelines for automatic validation.
- Tested system performance on real-world datasets (KITTI), achieving smooth and loop-free trajectory plots in 3D space.

WORK EXPERIENCE

Smart Internz, Hyderabad, India

Jul 2022 – Sep 2022

Salesforce Developer Intern

- Gained hands-on experience in **Salesforce fundamentals**, organizational setup, process automation, security, and API integration.
- Developed and debugged **Apex code** configured **Lightning Web Components (LWC)** and set up **VS Code & CLI** for development.
- Earned **Super badges: Apex Specialist & Process Automation Specialist**, demonstrating expertise in Salesforce automation and development.
- Participated in Agile standups and peer reviews while contributing to codebase improvements.

MALtech Solutions Pvt. Ltd, Bapatla, India.

Nov 2021 – Dec 2021

Web Developer

- Made web applications that are easy to expand using React.js, AngularJS, and Next.js, as well as RESTful APIs and additional libraries to enhance and improve the user experience.
- Implemented **responsive UI/UX components** with **HTML5, CSS3, Bootstrap, and Tailwind CSS**, ensuring optimal performance across devices and browsers.
- Redux was used to manage the state for the application more efficiently and boost the performance of the frontend when there were many details.
- Developed backend APIs using Node.js and Express, ensuring secure data handling.
- Deployed and managed applications using **AWS EC2, S3, and CloudFront**, incorporating **GitHub Actions** and **Docker** for CI/CD automation.

EDUCATION

Master's in computer science

Saint Louis University, Saint Louis, MO, USA

Aug 2023 - May 2025

Batcheler's in computer science

Chalapathi Institute of Engineering and Technology, Guntur, India.

Aug 2019 – May 2023